

AI-POWERED CUSTOMER SUPPORT ASSISTANT

Business Requirements Document

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Document Control

3.1 Revision History

Version	Date	Author	Change Summary
Stakeholder Role	Name	Sign-off Date	Status
Product Manager	Notional Stakeholder	—	Not Applicable — Self-Initiated Case Study
Support Manager	Notional Stakeholder	—	Not Applicable — Self-Initiated Case Study
Knowledge Management Lead	Notional Stakeholder	—	Not Applicable — Self-Initiated Case Study

1. Project Overview

1.1 Background

1.2 Problem Statement

1.3 Business Impact

1.4 Business Objectives

1.1 Background

CloudDesk is a SaaS-based customer service management platform used by businesses to manage customer support tickets, customer issues, support workflows, and related service operations.

CloudDesk supports its customers through a dedicated customer support team. Support agents assist customers with product-related questions, troubleshooting issues, account-related queries, and other common support requests.

To resolve these queries, support agents rely on multiple knowledge sources such as product documentation, FAQs, troubleshooting guides, and internal support articles. As the volume of documentation and customer queries increases, agents may face challenges in quickly locating the information required to provide accurate and timely responses.

This project proposes an AI-powered Customer Support Assistant to help CloudDesk support agents access relevant information more efficiently while keeping human agents responsible for verification and customer communication.

1.2 Problem Statement

Support agents currently spend significant time searching across multiple knowledge sources, including product documentation, FAQs, troubleshooting guides, and internal support articles, to find relevant information needed to resolve customer queries. As the knowledge base grows, locating accurate and relevant information becomes increasingly time-consuming, resulting in longer customer response and resolution times, increased workload for support agents, and inconsistent responses.

1.3 Business Impact

The current knowledge-search process can negatively impact support operations and customer experience. The key business impacts include:

- **Increased response and resolution time:** Support agents may take longer to respond to and resolve customer queries.
- **Higher support-agent workload:** Agents spend valuable time searching through multiple knowledge sources instead of focusing on customer interactions and complex issues.
- **Reduced productivity:** Repetitive knowledge-search activities can reduce overall support-team efficiency.
- **Inconsistent responses:** Difficulty locating the most relevant or updated information may result in inconsistent responses to similar customer queries.
- **Increased escalation:** Routine queries may be unnecessarily escalated to senior support teams when agents cannot quickly locate the required information.
- **Potential impact on customer satisfaction:** Longer resolution times and inconsistent responses may negatively affect the overall customer experience.

1.4 Business Objectives

The organization aims to improve the efficiency, consistency, and effectiveness of its customer support operations through better access to relevant knowledge.

ID	Business Objective
BO-001	Reduce time support agents spend searching across knowledge sources for relevant information.
BO-002	Improve speed of responding to customer queries.
BO-003	Improve consistency and accuracy of information provided to customers.
BO-004	Reduce unnecessary escalation of routine customer queries
BO-005	Improve overall support-agent productivity and customer experience.

2. Stakeholder Analysis

The project involves stakeholders across customer support, product, technology, knowledge management, security, and business operations. Each stakeholder has different responsibilities, expectations, and interests in the proposed AI Customer Support Assistant.

Stakeholder	Role	Interest / Need
Support Agent	Primary user	Quickly find accurate information to resolve customer queries
Support Manager	Business owner	Improve support efficiency, response time, and agent productivity
Product Manager	Product owner	Ensure the solution meets business and user needs
AI/Engineering Team	Technical team	Design, develop, integrate, and maintain the AI/RAG solution
Knowledge Management Team	Content owner	Maintain accurate, relevant, and up-to-date knowledge sources
Security & Compliance Team	Governance	Ensure customer and company information is handled securely
Business/Operations Team	Business stakeholder	Evaluate business impact, operational improvements, and costs
Customer	End beneficiary	Receive faster, accurate, and consistent support

3. Scope

3.1 In Scope

The proposed AI Customer Support Assistant will focus on helping support agents quickly find and understand relevant information from approved organizational knowledge sources.

The following capabilities are included within the scope of this project:

- Allowing authenticated support agents to submit natural-language questions.
- Retrieving relevant information from approved knowledge sources such as product documentation, FAQs, troubleshooting guides, and internal support articles.
- Generating AI-assisted responses using retrieved information.
- Displaying supporting source references for generated responses.
- Maintaining conversation context during an active interaction.
- Allowing support agents to provide feedback on AI-generated responses.
- Providing fallback guidance when relevant information cannot be retrieved.
- Providing an option to escalate queries to human or senior support when required.

3.2 Out of Scope

The following capabilities are outside the scope of the proposed solution:

- Direct autonomous communication with customers.
- Voice-based customer support.
- Automatic refunds, account changes, or other transactional actions.
- Replacement of the organization's existing CRM or ticketing system.
- Training or developing a new large language model from scratch.
- Making business decisions without human review.
- Retrieval from unapproved or unauthorized information sources.

3.3 Assumptions

The following assumptions have been made for the purpose of this self-initiated case study:

- The organization maintains existing product documentation, FAQs, troubleshooting guides, and internal support articles.
- Relevant knowledge sources can be made available to the proposed AI assistant.
- Support agents have authenticated access to the system.
- The available knowledge sources contain sufficient information to address a meaningful portion of routine support queries.
- Human support remains available for queries that cannot be reliably addressed by the AI assistant.

- AI-generated responses will be monitored and evaluated for accuracy, relevance, and groundedness.

3.4 Constraints

The proposed solution will operate within the following constraints:

- The solution will rely on existing or approved AI/LLM capabilities rather than developing a new LLM from scratch.
- The assistant should retrieve information only from approved organizational knowledge sources.
- Access to the solution and underlying information must comply with applicable security and privacy requirements.
- Human oversight should remain available for uncertain, sensitive, or complex queries.
- The initial solution scope is limited to assisting support agents and does not include autonomous customer communication.

3.5 Dependencies

The successful implementation of the proposed solution depends on:

- Availability of accurate and up-to-date organizational knowledge sources.
- Access to an appropriate LLM or AI service.
- Availability of retrieval and search infrastructure for processing organizational knowledge.
- Authentication and access-control mechanisms for support agents.
- Collaboration between product, engineering, support, knowledge management, and security teams.
- Ongoing maintenance and updating of the knowledge sources used by the assistant.

4. Business Requirements

The following business requirements define the capabilities needed to address the identified support challenges and achieve the project's business objectives.

ID	Business Requirement	Related Objective
BR-001	The organization should provide support agents with a centralized way to search approved knowledge sources using natural-language queries.	BO-001

BR-002	The solution should help support agents quickly identify relevant information required to resolve customer queries.	BO-001, BO-002
BR-003	The solution should provide AI-assisted responses based on information retrieved from approved knowledge sources.	BO-002, BO-003
BR-004	The solution should provide supporting source references so that support agents can verify the information used to generate an AI response.	BO-003
BR-005	The solution should help reduce unnecessary escalation of routine customer queries by providing relevant information to support agents.	BO-004
BR-006	The solution should allow support agents to provide feedback on the usefulness and quality of AI-generated responses.	BO-003, BO-005
BR-007	The solution should provide appropriate fallback or escalation guidance when sufficient information is not available to answer a query reliably.	BO-003, BO-004
BR-008	The solution should ensure that AI-assisted responses are generated using only approved and authorized knowledge sources.	BO-003, BO-005
BR-009	The organization should be able to evaluate the effectiveness of the AI assistant using defined performance and business metrics.	BO-005
BR-010	The solution should support human oversight for uncertain, sensitive, or complex customer queries	BO-003, BO-005
BR-011	The solution could provide proactive suggestions for relevant knowledge articles based on commonly occurring support query patterns..	BO-001, BO-005

4.1 Business Requirement Prioritization

Business requirements will be prioritized based on their importance to the initial solution.

Priority	Meaning
Must Have	Essential for the initial solution to achieve its primary business objectives
Should Have	Important capability that improves the solution but is not critical for initial implementation
Could Have	Valuable enhancement that can be considered in a future phase

Requirement ID	Priority
BR-001	Must Have
BR-002	Must Have
BR-003	Must Have
BR-004	Must Have
BR-005	Must Have
BR-006	Should Have
BR-007	Must Have
BR-008	Must Have
BR-009	Should Have
BR-010	Must Have
BR-011	Could have

4.2 Proposed Phasing

The proposed solution should be implemented in phases to validate the core support-assistance workflow before introducing more advanced capabilities.

Phase 1 — MVP

The initial MVP should focus on the capabilities required to provide safe, useful AI assistance to support agents.

MVP capabilities:

- Natural-language query submission
- Retrieval from approved knowledge sources
- AI-generated responses based on retrieved information
- Supporting source references
- Fallback when sufficient information is unavailable
- Human verification before customer communication
- Basic agent feedback
- Authentication and access control

These capabilities correspond primarily to the Must Have business requirements.

Phase 2 — Enhanced Support Experience

After the MVP has been evaluated, additional capabilities can be introduced to improve workflow efficiency and insight.

Phase 2 capabilities:

- Conversation context for follow-up queries
- Advanced feedback analysis
- Knowledge-gap identification and reporting
- Expanded monitoring and analytics
- Additional approved knowledge sources
- Improved support workflow integrations

These capabilities primarily build on Should Have requirements.

Future Enhancement

BR-011 — Proactive Knowledge Suggestions could be considered as a future enhancement once sufficient usage data is available to identify common query patterns and frequently referenced knowledge.

This phased approach allows the organization to validate response quality, agent usefulness, safety, and business impact before expanding the solution.

5. Functional Requirements

Functional requirements define the specific capabilities and behaviors that the proposed AI Customer Support Assistant should provide to support the identified business requirements.

5.1 Query Submission

ID	Functional Requirement	Related BR
FR-001	The system shall allow authenticated support agents to enter customer queries using natural language.	BR-001
FR-002	The system shall accept queries related to product usage, troubleshooting, FAQs, and other supported customer-support topics.	BR-001
FR-003	The system shall validate that a query contains sufficient input for processing and prompt the agent when the submitted query is empty or unclear.	BR-001

5.2 Knowledge Retrieval

ID	Functional Requirement	Related BR
FR-004	The system shall retrieve relevant information from approved organizational knowledge sources based on the submitted query.	BR-001, BR-002
FR-005	The system shall use relevant document content to identify information that can support the response.	BR-002, BR-003

FR-006	The system shall restrict retrieval to authorized and approved knowledge sources.	BR-008
FR-007	The system shall indicate when relevant information cannot be found in the available knowledge sources.	BR-007

5.3 AI Response Generation

ID	Functional Requirement	Related BR
FR-008	The system shall generate an AI-assisted response using the information retrieved from approved knowledge sources.	BR-003
FR-009	The system shall generate responses that are relevant to the support agent's submitted query.	BR-002, BR-003
FR-010	The system shall avoid presenting unsupported information as factual when sufficient supporting information is not available.	BR-003, BR-007
FR-011	The system shall maintain relevant conversation context during an active interaction to support follow-up questions.	BR-003
FR-012	The system shall provide a clear indication when the response requires verification or human review.	BR-007, BR-010

5.4 Source References

ID	Functional Requirement	Related BR
FR-013	The system shall display the source or sources used to generate an AI-assisted response.	BR-004
FR-014	The system shall allow support agents to review the referenced source information before using the response with a customer.	BR-004
FR-015	The system shall identify when no suitable source is available for a generated response.	BR-004, BR-007

5.5 Agent Feedback

ID	Functional Requirement	Related BR
FR-016	The system shall allow support agents to provide feedback on AI-generated responses.	BR-006
FR-017	The system shall support basic feedback options such as helpful and not helpful.	BR-006
FR-018	The system should allow agents to provide additional feedback when a response is inaccurate, irrelevant, or incomplete.	BR-006

5.6 Fallback and Escalation

ID	Functional Requirement	Related BR
FR-019	The system shall provide fallback guidance when relevant information cannot be reliably retrieved.	BR-007
FR-020	The system shall recommend escalation to human or senior support when a query cannot be reliably addressed by the assistant.	BR-007, BR-010
FR-021	The system shall not provide autonomous transactional actions such as refunds, account changes, or other customer-impacting actions.	BR-010

5.7 Access and Security

ID	Functional Requirement	Related BR
FR-022	The system shall require authenticated access for support agents.	BR-008
FR-023	The system shall ensure that users can access only information they are authorized to view.	BR-008
FR-024	The system shall prevent retrieval from unapproved or unauthorized knowledge sources.	BR-008

5.8 Monitoring and Evaluation

ID	Functional Requirement	Related BR
FR-025	The system shall capture relevant feedback and usage information required to evaluate assistant performance.	BR-006, BR-009
FR-026	The organization should be able to evaluate response quality using defined AI and business performance metrics.	BR-009
FR-027	The system should support monitoring of cases where the assistant fails to provide a reliable answer or requires human escalation.	BR-007, BR-009

6. Non-Functional Requirements

Non-functional requirements define the expected quality, performance, security, reliability, and usability characteristics of the proposed AI Customer Support Assistant.

6.1 Performance

ID	Non-Functional Requirement	Related Requirement
NFR-001	The system should provide an AI-assisted response within an acceptable response time for normal support queries.	FR-008
NFR-002	The system should process user queries without unnecessary delays during normal operating conditions.	FR-001
NFR-003	The system should maintain consistent performance when multiple support agents use the assistant simultaneously.	FR-001, FR-008

6.2 Accuracy and Response Quality

ID	Non-Functional Requirement	Related Requirement
NFR-004	AI-generated responses should be relevant to the submitted support query.	FR-009
NFR-005	AI-generated responses should be grounded in information retrieved from approved knowledge sources.	FR-008, FR-010
NFR-006	The system should minimize unsupported or fabricated information in AI-generated responses.	FR-010
NFR-007	The quality of AI responses should be evaluated using defined evaluation criteria and performance metrics.	FR-026

6.3 Security and Access Control

ID	Non-Functional Requirement	Related Requirement
NFR-008	The system shall restrict access to authenticated and authorized support agents.	FR-022
NFR-009	The system shall protect organizational and customer-related information from unauthorized access.	FR-023
NFR-010	The system shall ensure that retrieval is limited to approved knowledge sources and authorized information.	FR-024

6.4 Reliability and Availability

ID	Non-Functional Requirement	Related Requirement
NFR-011	The system should be available during normal support operating hours.	FR-001
NFR-012	The system should handle temporary failures in retrieval or AI response generation gracefully.	FR-007, FR-019
NFR-013	The system should provide an appropriate fallback or escalation path when the AI assistant is unavailable.	FR-019, FR-020

6.5 Usability

ID	Non-Functional Requirement	Related Requirement
NFR-014	The interface should be simple and intuitive for support agents with minimal training.	FR-001
NFR-015	AI-generated responses and source references should be presented in a clear and understandable format.	FR-008, FR-013
NFR-016	The system should make it easy for support agents to review, verify, and use AI-assisted responses.	FR-014

6.6 Scalability

ID	Non-Functional Requirement	Related Requirement
NFR-017	The solution should be capable of supporting an increasing number of support agents and queries without significant degradation in performance.	FR-001, FR-008
NFR-018	The solution should allow additional approved knowledge sources to be incorporated as organizational requirements evolve.	FR-004, FR-006

6.7 Maintainability and Knowledge Management

ID	Non-Functional Requirement	Requirement
NFR-019	The solution should support the use of updated knowledge sources so that responses remain aligned with current product information.	FR-004
NFR-020	Changes to approved knowledge sources should be reflected in the retrieval process within an appropriate operational timeframe.	FR-004, FR-006
NFR-021	The solution should allow monitoring and evaluation processes to identify recurring response-quality issues.	FR-025, FR-027

6.8 Human Oversight

ID	Non-Functional Requirement	Related Requirement
NFR-022	The solution should maintain human oversight for uncertain, sensitive, or complex support queries.	FR-012, FR-020
NFR-023	Support agents should be able to verify AI-generated information before communicating it to customers.	FR-014
NFR-024	Query logs, agent feedback, and conversation context should be retained only for the period necessary for operational monitoring, evaluation, and improvement, in accordance with applicable organizational data-retention and privacy requirements.	FR-025, FR-026, FR-027

7. Risks and Mitigation

The proposed AI Customer Support Assistant introduces potential business, technical, operational, and AI-related risks. Identifying and mitigating these risks is important to ensure that the solution remains reliable, secure, and useful for support agents.

ID	Risk	Impact	Mitigation	Owner	Status
R-001	Inaccurate or unsupported AI information	Incorrect customer information and reduced support quality	RAG, source references, human verification	Product Manager / AI Engineering	Open
R-002	Relevant information is not available in approved sources	Assistant cannot provide a useful answer	Fallback and escalation to human support	Knowledge Management / Support Manager	Open
R-003	Outdated or incorrect knowledge sources	Responses may contain obsolete information	Knowledge ownership, review, and update process	Knowledge Management Team	Open
R-004	AI response is relevant but incomplete	Agent may assume the response fully addresses the issue	Source references and agent verification	Support Manager	Open
R-005	Unauthorized information retrieval	Potential exposure of sensitive information	Authentication, authorization, and retrieval restrictions	Security & Compliance / AI Engineering	Open
R-006	Overdependence on AI	Agents may accept incorrect information without sufficient review	Human oversight and verification	Support Manager	Open

R-007	AI or retrieval infrastructure unavailable	Assistant may become temporarily unusable	Existing support process and fallback path	AI/Engineering Team	Open
R-008	Poorly maintained knowledge sources	Declining retrieval and response quality	Knowledge monitoring and source maintenance	Knowledge Management Team	Open
R-009	Inconsistent AI quality across query types	Assistant may be less useful for certain support scenarios	Evaluation criteria, monitoring, and feedback analysis	Product Manager / AI Engineering	Open
R-010	Sensitive or complex queries are mishandled	Customer, operational, or compliance risk	Defined escalation criteria and human support	Support Manager / Security & Compliance	Open

7.1 Risk Prioritization

Risks will be prioritized based on their potential impact on customers, support operations, information security, and the reliability of the AI assistant.

Priority	Risk IDs	Priority Rationale
High	R-001, R-005, R-010	These risks could directly affect response accuracy, information security, or safe handling of sensitive queries.
Medium	R-003, R-004, R-006, R-008, R-009	These risks could negatively affect response quality, agent productivity, or trust in the solution.
Low	R-002, R-007	These risks can affect usability or availability but can be addressed through fallback processes and existing support procedures.

8. User Personas

User personas represent the primary users and stakeholders who interact with or are affected by the proposed AI Customer Support Assistant. The personas help define user needs, pain points, goals, and expectations that should be considered while designing the solution.

8.1 Primary User — Support Agent

Attribute	Description
Persona	Support Agent
Role	Handles customer queries and provides product-related support
Primary Goal	Resolve customer queries accurately and efficiently
Current Pain Points	Spends time searching across multiple documentation sources; may have difficulty identifying the most relevant information; may need to escalate routine queries when information is difficult to locate
Needs	Fast access to relevant information, clear AI-assisted responses, source references, and an easy way to verify information
Expected Benefit	Reduced knowledge-search time, faster responses, improved consistency, and increased productivity
Key Concern	Receiving inaccurate, outdated, or unsupported AI-generated information

8.2 Business Stakeholder — Support Manager

Attribute	Description
Persona	Support Manager
Role	Manages support operations, team performance, and service quality
Primary Goal	Improve support-team efficiency and customer experience
Current Pain Points	High agent workload, longer response times, unnecessary escalations, and inconsistent responses
Needs	Improved agent productivity, measurable performance improvements, and visibility into AI assistant effectiveness
Expected Benefit	Reduced response time, lower unnecessary escalation, and improved support-team efficiency

Key Concern Poor AI performance, low agent adoption, and potential negative impact on customer experience

8.3 Supporting Stakeholder — Knowledge Management Team

Attribute	Description
Persona	Knowledge Management Specialist
Role	Maintains and manages product documentation, FAQs, troubleshooting guides, and internal support content
Primary Goal	Ensure support agents and the AI assistant have access to accurate and up-to-date information
Current Pain Points	Outdated or inconsistent documentation may reduce the usefulness of support knowledge
Needs	Clear ownership of knowledge sources, content maintenance processes, and visibility into recurring knowledge gaps
Expected Benefit	Better utilization of organizational knowledge and improved response quality
Key Concern	Outdated, incomplete, or conflicting information being retrieved by the assistant

Attribute	Description
Persona	AI/Engineering Team
Role	Designs, develops, integrates, and maintains the technical solution
Primary Goal	Build a reliable and maintainable AI assistant that meets defined business and functional requirements
Current Pain Points	Need to integrate AI capabilities with organizational knowledge sources while maintaining performance, security, and reliability
Needs	Clear requirements, defined system behavior, security constraints, evaluation criteria, and measurable success metrics
Expected Benefit	Clear product requirements and well-defined expectations for implementation

Key Concern	Integration complexity, system reliability, AI response quality, and security requirements
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9. User Journey

The user journey describes how a support agent interacts with the existing knowledge-search process and how the proposed AI Customer Support Assistant would improve that experience.

9.1 Current-State User Journey

Step	Support Agent Activity	Pain Point
1	Receives a customer query.	Agent needs to understand the customer's issue before searching for information.
2	Identifies the type of information required.	It may not be immediately clear which knowledge source contains the answer.
3	Searches product documentation, FAQs, troubleshooting guides, or internal support articles.	Agent may need to search across multiple sources.
4	Reviews multiple search results or documents.	Finding the most relevant information can take time.
5	Determines whether the information is sufficient to answer the customer.	Information may be incomplete, outdated, or difficult to interpret.
6	Provides a response to the customer or escalates the issue.	Routine queries may be unnecessarily escalated when information is difficult to locate.

9.2 Future-State User Journey

Step	Support Agent Activity	AI Assistant Activity	Expected Benefit
1	Receives a customer query.	—	Customer issue is identified.

2	Enters the query into the AI Customer Support Assistant.	Accepts and processes the natural-language query.	Reduces manual searching effort.
3	Reviews the AI-assisted response.	Retrieves relevant information from approved knowledge sources.	Provides faster access to relevant information.
4	Verifies the response and source references.	Generates a response grounded in retrieved information and displays supporting sources.	Improves confidence and consistency.
5	Uses the verified information to respond to the customer.	—	Enables faster customer responses.
6	Provides feedback or requests further assistance when needed.	Captures feedback and maintains conversation context where applicable.	Supports continuous improvement.
7	Escalates the query when the AI cannot reliably answer it.	Provides fallback or escalation guidance.	Prevents over-reliance on unsupported AI responses.

9.3 Key Improvements in the Future-State Journey

The proposed journey is intended to improve the support process in the following areas:

- **Reduced knowledge-search effort:** Agents can submit a natural-language query instead of manually searching multiple sources.

- **Faster access to relevant information:** The assistant retrieves information from approved knowledge sources.
- **Improved response consistency:** Agents receive AI-assisted responses based on organizational knowledge.
- **Better verification:** Source references allow agents to review the information supporting an AI-generated response.
- **Safer handling of uncertainty:** The assistant provides fallback and escalation guidance when reliable information is unavailable.
- **Continuous improvement:** Agent feedback can be used to identify recurring response-quality and knowledge gaps.

10. User Stories

User stories describe the needs and expected outcomes of users from their perspective. They are written in a simple format to communicate what the user wants to accomplish and why.

The primary user for the proposed AI Customer Support Assistant is the Support Agent.

10.1 Support Agent User Stories

ID	User Story	Related Requirement	Priority
US-001	As a support agent, I want to enter a customer query in natural language so that I can quickly search for relevant information.	FR-001	Must Have
US-002	As a support agent, I want the assistant to retrieve relevant information from approved	FR-004	Must Have

knowledge sources so that I do not have to search multiple sources manually.

US -00 3	As a support agent, I want the assistant to generate an AI-assisted response based on retrieved information so that I can respond to customers more efficiently.	FR-008	Must Have
US -00 4	As a support agent, I want to see the sources used for an AI-generated response so that I can verify the information before communicating it to the customer.	FR-013, FR-014	Must Have
US -00 5	As a support agent, I want the assistant to maintain conversation context so that I can ask follow-up questions without repeating the entire issue.	FR-011	Should Have
US -00 6	As a support agent, I want the assistant to indicate when it does not have sufficient information so that I do not rely on an unsupported answer.	FR-007, FR-010	Must Have
US -00 7	As a support agent, I want clear escalation guidance when the assistant cannot reliably answer a query so that complex issues can be handled by human support.	FR-019, FR-020	Must Have
US -00 8	As a support agent, I want to provide feedback on AI-generated responses so that response quality can be monitored and improved.	FR-016, FR-017	Should Have
US -00 9	As a support agent, I want to know when a response requires additional verification so that I can review it before sharing information with the customer.	FR-012	Must Have
US -01 0	As a support agent, I want to access only information that I am authorized to view so that sensitive organizational information remains protected.	FR-022, FR-023	Must Have

10.2 Support Manager User Stories

ID	User Story	Related Requirement	Priority
US-011	As a support manager, I want to monitor the assistant's usage and response quality so that I can evaluate whether it is improving support operations.	FR-025, FR-026	Should Have
US-012	As a support manager, I want to identify queries that frequently require escalation so that recurring support or knowledge gaps can be addressed.	FR-027	Should Have
US-013	As a support manager, I want to understand agent feedback on AI responses so that I can identify opportunities to improve the assistant.	FR-016, FR-025	Should Have

10.3 Knowledge Management User Stories

ID	User Story	Related Requirement	Priority
US-014	As a Knowledge Management Team member, I want to manage the set of approved knowledge sources so that the assistant retrieves information only from authorized content.	FR-006	Must Have
US-015	As a knowledge management specialist, I want to identify recurring knowledge gaps so that documentation can be improved or updated.	FR-025, FR-027	Should Have

10.4 AI/Engineering User Stories

ID	User Story	Related Requirement	Priority
US-016	As an AI/Engineering team member, I want clearly defined functional, non-functional, security, evaluation, and fallback requirements so that the proposed solution can be designed consistently with business expectations.	FR-001 to FR-027	Must Have
US-017	As an AI/engineering team member, I want the system to provide appropriate fallback behavior when retrieval or AI services fail so that support operations can continue.	FR-019, FR-020	Must Have

11. Acceptance Criteria

Acceptance criteria define the conditions that must be satisfied for a user story to be considered complete and acceptable. They provide clear, testable expectations for the proposed AI Customer Support Assistant and help ensure alignment between business, product, and technical stakeholders.

User Story ID	Acceptance Criteria
US-001	The support agent can enter a query using natural language. The system accepts valid queries and initiates processing.
US-002	The assistant retrieves relevant information only from approved knowledge sources. Retrieved information must be relevant to the submitted query.
US-003	The assistant generates a response using the information retrieved from approved sources. The response should directly address the agent's query.
US-004	The assistant displays the source or sources used to generate the response. The support agent can review the referenced source before using the response.

- US-005 The assistant maintains relevant conversation context during the active interaction so that follow-up questions can be understood in context.
- US-006 When sufficient supporting information cannot be found, the assistant clearly indicates that it cannot reliably answer the query instead of presenting unsupported information as fact.
- US-007 When the query cannot be reliably addressed by the assistant, the system provides clear guidance to escalate the query to human or senior support.
- US-008 The support agent can provide feedback on an AI-generated response using a helpful/not-helpful option. Additional feedback can be provided for inaccurate, irrelevant, or incomplete responses.
- US-009 The assistant clearly indicates when a response requires verification or human review before being used in customer communication.
- US-010 Only authenticated and authorized support agents can access the assistant and information they are permitted to view. Unauthorized knowledge sources or information must not be retrieved.
- US-011 The support manager can access relevant usage and response-quality information to evaluate the assistant's effectiveness.
- US-012 The system captures information that allows the support manager to identify query types or situations that frequently result in escalation.
- US-013 The support manager can review aggregated agent feedback to identify recurring issues with AI response quality or usefulness.
- US-014 The knowledge management team can identify and maintain the approved knowledge sources used by the assistant. The assistant must not retrieve information from unapproved sources.
- US-015 The system provides information that helps the knowledge management team identify recurring knowledge gaps or areas where available documentation may be insufficient.
- US-016 The AI/Engineering team has access to defined functional, non-functional, security, evaluation, and fallback requirements needed to design and maintain the solution.
- US-017 If the retrieval or AI service is unavailable or fails, the system provides an appropriate fallback or escalation path rather than returning an unsupported response.

11.1 Acceptance Criteria Principles

The acceptance criteria for the proposed solution are designed around the following principles:

- **Testability:** Each criterion describes an observable system behavior or outcome.
- **Accuracy and grounding:** AI responses should be based on approved and retrieved information.
- **Transparency:** Source references should allow support agents to verify AI-generated responses.
- **Human oversight:** Agents remain responsible for reviewing information before communicating with customers.
- **Safe failure:** The assistant should acknowledge uncertainty and provide fallback or escalation guidance rather than generate unsupported information.
- **Access control:** Users should only retrieve information they are authorized to access.
- **Continuous improvement:** Feedback, usage data, and recurring failure patterns should support ongoing evaluation and improvement of the solution.

12. Proposed Solution

12.1 Solution Overview

The proposed solution is an AI-powered Customer Support Assistant designed to help CloudDesk support agents quickly find relevant and reliable information while resolving customer queries.

The assistant will allow authenticated support agents to submit customer-related questions in natural language. The system will retrieve relevant information from approved knowledge sources such as product documentation, FAQs, troubleshooting guides, and internal support articles. An AI language model will then generate a response based on the retrieved information and provide supporting source references for agent verification.

The solution is intended to **assist support agents rather than replace human decision-making or customer support interactions**. Support agents remain responsible for reviewing and validating AI-generated information before communicating it to customers.

12.2 Proposed Solution Workflow

The proposed solution will follow the workflow below:

1. **Agent receives a customer query**
The support agent receives a customer question or issue through the existing support process.
2. **Agent submits the query**
The agent enters the customer query into the AI Customer Support Assistant using natural language.
3. **Query processing**
The assistant processes the query and identifies the information required to address it.
4. **Knowledge retrieval**
The system searches approved and authorized knowledge sources for relevant information.
5. **Relevant information is retrieved**
The most relevant information is selected and provided to the AI response-generation component.
6. **AI response generation**
The AI generates a response based on the retrieved information rather than relying solely on its general language knowledge.
7. **Source references are displayed**
The assistant displays the source documents or references used to support the response.
8. **Agent reviews the response**
The support agent reviews the generated response and verifies the supporting information.
9. **Agent responds or escalates**
If the information is sufficient, the agent uses the verified information to respond to the customer. If the assistant cannot reliably address the query, the agent follows the recommended escalation path.
10. **Feedback is captured**
The agent can provide feedback on whether the response was useful, accurate, relevant, or complete. This feedback can support future evaluation and improvement.

12.3 Key Solution Components

The proposed solution consists of the following major components:

Component	Purpose
Agent Interface	Allows authenticated support agents to submit queries, view responses, review sources, and provide feedback.
Query Processing	Processes the agent's natural-language query before information retrieval.
Knowledge Retrieval	Searches approved knowledge sources for relevant information.
Approved Knowledge Base	Contains product documentation, FAQs, troubleshooting guides, and internal support articles that can be used by the assistant.
AI Response Generation	Generates an AI-assisted response using the retrieved information.
Source References	Shows the supporting knowledge sources used for the generated response.
Feedback Mechanism	Allows agents to provide feedback on response usefulness and quality.
Fallback & Escalation	Handles situations where sufficient or reliable information cannot be retrieved.
Monitoring & Evaluation	Captures relevant usage, feedback, and response-quality information for evaluating the solution.
Access & Security Controls	Ensures that only authenticated users can access information they are authorized to view.

12.4 Proposed AI Approach

The solution will use a **Retrieval-Augmented Generation (RAG)** approach.

Instead of relying only on the AI model's pre-existing knowledge, the assistant will first retrieve relevant information from CloudDesk's approved knowledge sources. The retrieved information will then be provided as context to the AI model to generate a response.

This approach is proposed to improve the relevance and grounding of responses and reduce the risk of unsupported information being presented as fact.

The detailed RAG architecture and workflow are described in the subsequent sections of this document.

12.5 Human-in-the-Loop Approach

Human oversight is a core part of the proposed solution.

The AI assistant will provide recommendations and information to support agents, but it will not independently communicate with customers or make customer-impacting decisions.

Support agents will:

- Review AI-generated responses.
- Verify supporting source references.
- Determine whether the information is appropriate for the customer query.
- Escalate queries when the assistant cannot reliably provide an answer.
- Provide feedback on response quality.

This approach is intended to balance the efficiency benefits of AI assistance with the need for human judgment and accountability.

12.6 Expected Business Benefits

If successfully implemented, the proposed solution is expected to support the following business outcomes:

- Reduce the time agents spend searching across multiple knowledge sources.
- Improve the speed of responding to routine customer queries.
- Improve consistency in information provided to customers.
- Reduce unnecessary escalation of routine queries.
- Improve support-agent productivity.
- Provide greater visibility into recurring knowledge gaps and AI response-quality issues.
- Create a feedback loop for continuous improvement of the support-assistance experience.

13. RAG Architecture

13.1 Overview

The proposed AI Customer Support Assistant will use a **Retrieval-Augmented Generation (RAG)** architecture to generate responses based on CloudDesk's approved knowledge sources.

RAG combines information retrieval with AI-generated responses. Instead of relying only on the language model's existing knowledge, the system retrieves relevant information from the organization's knowledge base and provides that information to the language model as context for generating the response.

The proposed architecture is designed to improve response relevance, provide source transparency, and reduce the risk of unsupported information being presented as factual.

13.2 High-Level Architecture

The proposed solution consists of the following layers:

Support Agent → Assistant Interface → Query Processing → Knowledge Retrieval → Retrieved Context → AI/LLM → Response + Sources → Agent Verification

The knowledge sources form the foundation of the retrieval layer:

Product Documentation + FAQs + Troubleshooting Guides + Internal Support Articles → Knowledge Processing → Searchable Knowledge Base

13.3 Architecture Components

Component	Description
Support Agent	The primary user who submits customer-related queries and reviews AI-generated responses.
Assistant Interface	The interface through which support agents submit queries, view responses, review sources, and provide feedback.
Query Processing	Processes the agent's natural-language query so that relevant information can be retrieved.
Approved Knowledge Sources	Product documentation, FAQs, troubleshooting guides, and approved internal support articles.
Document Processing	Processes knowledge documents so that their content can be prepared for retrieval.

Embedding Model	Converts document content and user queries into numerical representations that can be used to identify semantically relevant information.
Vector Database / Search Index	Stores searchable representations of approved knowledge content and enables relevant information to be retrieved.
Retrieval Layer	Searches the approved knowledge base and returns information relevant to the submitted query.
Context Builder	Organizes retrieved information into context that can be provided to the AI model.
LLM / Generative AI Model	Generates a natural-language response using the user's query and retrieved information.
Source Reference Layer	Provides references to the knowledge sources used to support the generated response.
Feedback Mechanism	Captures agent feedback on the usefulness and quality of responses.
Monitoring & Evaluation	Tracks relevant usage, feedback, response quality, retrieval performance, and escalation patterns.
Access & Security Controls	Ensures users and retrieved information are subject to appropriate authentication and authorization controls.
Fallback & Escalation	Handles situations where relevant information cannot be reliably retrieved or the AI cannot provide a suitable response.

13.4 Knowledge Ingestion Flow

Before support agents can use the assistant, approved knowledge sources must be processed and made searchable.

The proposed flow is:

Approved Documents → Document Processing → Content Chunking → Embeddings → Vector Database / Search Index

1. Approved documents are collected from authorized knowledge sources.
2. Documents are processed to extract usable content.
3. Large documents are divided into smaller sections or **chunks**.

4. Each chunk is converted into an embedding, which represents its semantic meaning in numerical form.
5. The embeddings and relevant document metadata are stored in a vector database or search index.
6. The resulting knowledge base becomes available to the retrieval component.

Only approved and authorized knowledge sources should be included in this process.

13.5 Query and Response Flow

When a support agent submits a question, the proposed system follows this flow:

Agent Query → Query Processing → Retrieval → Relevant Context → LLM → Grounded Response → Source References → Agent Verification

1. The support agent enters a natural-language query.
2. The system processes the query.
3. The retrieval layer searches the approved knowledge base.
4. Relevant content is retrieved based on the query.
5. The retrieved content is provided to the LLM as contextual information.
6. The LLM generates a response using the query and retrieved context.
7. Supporting source references are displayed with the response.
8. The agent reviews the response and supporting sources.
9. The agent either uses the verified information to respond to the customer or follows the escalation process.

13.6 Grounding and Response Safety

The AI response should be grounded in the information retrieved from approved knowledge sources.

When the retrieved information does not sufficiently support an answer, the assistant should not present an unsupported answer as fact. Instead, it should indicate that sufficient information could not be found and guide the agent toward verification or escalation.

This design supports the following principles:

- Responses should be based on approved organizational knowledge.
- Supporting sources should be visible to the agent.
- Uncertainty should be communicated clearly.
- Unsupported information should not be presented as confirmed information.
- Human review should remain part of the customer-response process.

13.7 Security and Access Considerations

The architecture should incorporate access controls throughout the retrieval process.

The system should ensure that:

- Only authenticated support agents can access the assistant.
- Agents can retrieve only information they are authorized to access.
- Unapproved knowledge sources are excluded from the searchable knowledge base.
- Access permissions are considered when retrieving information.
- Sensitive organizational or customer information is protected from unauthorized access.

13.8 Fallback Path

If the retrieval or generation process cannot reliably produce a useful response, the system should follow a safe fallback path:

Query → Retrieval → Insufficient/Unreliable Information → Uncertainty Message → Human Verification/Escalation

The assistant should not attempt to provide a confident answer when sufficient supporting information is unavailable.

The support agent can then use existing knowledge sources or escalate the issue to an appropriate human support team.

14. RAG Workflow

14.1 Overview

The Retrieval-Augmented Generation (RAG) workflow describes how the AI Customer Support Assistant processes a support agent's query, retrieves relevant information from approved knowledge sources, and generates a grounded response.

The workflow is designed to ensure that AI-generated responses are supported by relevant organizational knowledge and can be verified by the support agent before being used in customer communication.

14.2 Knowledge Preparation Workflow

Before queries can be processed, approved knowledge sources must be prepared for retrieval.

Workflow:

Approved Knowledge Sources → Document Processing → Content Chunking → Embedding Generation → Vector Database / Search Index

1. **Identify approved sources**
Product documentation, FAQs, troubleshooting guides, and approved internal support articles are identified as knowledge sources.
2. **Process documents**
The content of the approved documents is extracted and prepared for processing.
3. **Chunk the content**
Large documents are divided into smaller sections so that relevant information can be retrieved more effectively.
4. **Generate embeddings**
Each content chunk is converted into a numerical representation called an embedding.
5. **Store searchable content**
The embeddings, along with relevant metadata such as document name and source information, are stored in a vector database or search index.
6. **Maintain the knowledge base**
Approved knowledge sources should be reviewed and updated when the underlying product information changes.

14.3 Query Processing Workflow

When a support agent submits a query, the following workflow is followed:

Agent Query → Query Processing → Retrieval → Relevance Check → Context Preparation → AI Generation → Source References → Agent Verification

Step 1: Query Submission

The support agent enters a customer-related question into the AI Customer Support Assistant using natural language.

Example:

"How can a customer reset their account password?"

The system accepts the query and begins processing.

Step 2: Query Processing

The system processes the query to identify the information required to answer it.

The query may also be converted into an embedding so that semantically relevant content can be identified during retrieval.

Step 3: Knowledge Retrieval

The retrieval component searches the approved knowledge base for information relevant to the query.

The search should be restricted to knowledge sources and information that the agent is authorized to access.

Step 4: Relevance Check

The system evaluates whether the retrieved information is sufficiently relevant to the query.

Two broad outcomes are possible:

Relevant information found → Continue to response generation

Insufficient or irrelevant information → Fallback / escalation

This step helps prevent the AI model from generating an answer when the knowledge base does not contain adequate supporting information.

Step 5: Context Preparation

Relevant retrieved content is organized into context that can be provided to the AI model along with the original query.

The context should contain the information necessary for generating the response while retaining source information for later verification.

Step 6: AI Response Generation

The LLM receives:

- The original support query.
- Relevant retrieved information.
- Instructions governing how the response should be generated.

The model generates an AI-assisted response based on the provided context.

Step 7: Source References

The assistant displays the sources associated with the retrieved information used to generate the response.

This allows the support agent to understand where the information came from and verify it before communicating with the customer.

Step 8: Agent Verification

The support agent reviews:

- The AI-generated response.
- The supporting source references.
- Whether the response adequately addresses the customer's issue.

The agent remains responsible for deciding whether the response is appropriate for customer communication.

Step 9: Response or Escalation

If the response is sufficiently supported and appropriate, the agent can use the verified information to respond to the customer.

If the assistant cannot reliably address the query, the agent follows the appropriate fallback or escalation process.

Step 10: Feedback and Monitoring

The agent can provide feedback on the response, such as whether it was helpful or not helpful.

Relevant feedback and usage information can be used to evaluate:

- Response quality.
- Retrieval effectiveness.
- Frequently unanswered questions.
- Escalation patterns.
- Knowledge gaps.
- Recurring response issues.

14.4 End-to-End RAG Workflow

The complete workflow can be represented as:

1. Agent submits query

↓

2. Query is processed

↓

3. Approved knowledge base is searched

↓

4. Relevant information is retrieved

↓

5. Relevance/sufficiency is evaluated

↓

6. Relevant context is provided to the LLM

↓

7. AI generates a grounded response



8. Supporting sources are displayed



9. Agent verifies the response



10. Agent responds to customer OR escalates



11. Feedback and performance information are captured

14.5 Insufficient Information Workflow

If the system cannot find sufficient information to reliably answer the query:

Agent Query → Retrieval → Insufficient Information → Uncertainty Message → Human Verification / Existing Support Process → Escalation if Required

The assistant should clearly communicate that sufficient supporting information was not found rather than generating an unsupported response.

This fallback mechanism is important for maintaining trust and reducing the risk of inaccurate information being used in customer interactions.

14.6 Follow-Up Query Workflow

The assistant should maintain relevant context during an active conversation so that follow-up questions can be interpreted correctly.

For example:

Agent: "How do I reset a customer's password?"

Assistant: Provides password-reset guidance and source reference.

Agent: "What if the customer doesn't receive the reset email?"

The assistant should use the context of the previous interaction together with the new query to retrieve and provide relevant information.

14.7 Feedback Loop

Feedback from support agents can contribute to continuous improvement.

Agent Interaction → Feedback → Performance Analysis → Identify Issues / Knowledge Gaps → Improve Knowledge Sources or Solution → Re-evaluate

Feedback should not automatically be treated as proof that an AI response was factually correct or incorrect. Instead, it should be considered one input into ongoing evaluation and improvement.

14.8 Key Workflow Principles

The proposed RAG workflow follows these principles:

- **Grounded responses:** Responses should be based on retrieved approved information.
- **Source transparency:** Agents should be able to identify and review supporting sources.
- **Controlled access:** Retrieval should respect authentication and authorization requirements.
- **Safe failure:** The system should acknowledge insufficient information rather than fabricate an answer.
- **Human verification:** Agents remain responsible for reviewing information before customer communication.
- **Continuous improvement:** Feedback and monitoring should support identification of recurring issues and knowledge gaps.

15. Edge Cases & Fallback Scenarios

15.1 Overview

The AI Customer Support Assistant should be designed to handle situations where information is unavailable, ambiguous, conflicting, outdated, or where an AI or retrieval component fails.

The assistant should prioritize reliable information and safe handling of uncertainty over generating an answer for every query.

15.2 Edge Cases

ID	Scenario	Expected System Behavior	Fallback / Action
EC-001	No relevant information is found in the knowledge base	The assistant indicates that sufficient information could not be found.	Agent uses existing support resources or escalates the query.
EC-002	Retrieved information is insufficient to answer the complete query	The assistant identifies the limitation instead of presenting an incomplete answer as definitive.	Agent verifies available information and escalates if required.
EC-003	Multiple sources contain conflicting information	The assistant should not arbitrarily select one conflicting answer as definitive.	Flag the conflict for agent verification and knowledge-management review.
EC-004	Knowledge source contains outdated information	The assistant may retrieve the available information but should rely on source metadata/version information where available.	Agent verifies the information; outdated content is reviewed by the knowledge-management team.
EC-005	User submits an ambiguous query	The assistant requests clarification when the query does not provide enough information to identify the appropriate answer.	Agent provides additional context or asks the customer for clarification.

EC-006	Query is outside the supported product/support scope	The assistant identifies that the query is outside its supported scope.	Guide the agent toward the appropriate support process or team.
EC-007	Retrieval service fails	The assistant cannot retrieve knowledge-base information.	Display an appropriate service/fallback message and direct the agent to existing support resources or escalation.
EC-008	AI generation service fails	The system cannot generate an AI response.	Agent continues using existing support processes or knowledge sources.
EC-009	AI response is not adequately supported by retrieved information	The system should avoid presenting the response as a reliable answer.	Request verification or recommend escalation.
EC-010	Agent asks a follow-up question with insufficient context	The assistant requests clarification rather than assuming the intended meaning.	Agent provides additional context.
EC-011	Agent does not have authorization to access a relevant source	The restricted information should not be returned.	The agent follows an authorized support process or escalates appropriately.
EC-012	Sensitive or high-risk customer query is submitted	The assistant should avoid making unsupported decisions or providing unauthorized guidance.	Direct the query to the appropriate human support process.
EC-013	Knowledge source is temporarily unavailable	The retrieval system cannot access the affected source.	Use other approved available sources where appropriate or follow fallback/escalation.
EC-014	Agent provides negative feedback on a response	The feedback is captured for evaluation and analysis.	Investigate whether the issue relates to retrieval, source quality, response generation, or user experience.

EC-015	Same type of query repeatedly receives poor or incomplete responses	Recurring response-quality issues are identified through monitoring and feedback.	Review knowledge coverage, retrieval performance, and response-generation behavior.
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15.3 Fallback Principles

The fallback strategy should follow these principles:

1. Do not fabricate information

When sufficient supporting information is unavailable, the assistant should communicate the limitation rather than generate an unsupported answer.

2. Prefer clarification over assumption

When a query is ambiguous, the assistant should request additional information rather than assume the user's intended meaning.

3. Maintain human oversight

Queries that are uncertain, sensitive, complex, or outside the assistant's supported scope should be reviewed or handled through an appropriate human support process.

4. Protect restricted information

If an agent is not authorized to access a source or information, the assistant should not expose it merely because it could technically be retrieved.

5. Preserve existing support processes

Failure of the AI assistant should not prevent agents from continuing customer support through existing knowledge sources and support workflows.

6. Use failures as improvement signals

Repeated failures, escalations, negative feedback, and knowledge gaps should be monitored to identify opportunities to improve the knowledge base or AI solution.

15.4 Fallback Decision Flow

The proposed fallback decision process is:

Agent submits query

↓

Can relevant information be retrieved?

YES → Evaluate whether the information is sufficient

↓

Sufficient and reliable?

YES → Generate grounded response → Display sources → Agent verifies → Respond to customer

NO → Communicate uncertainty → Agent verifies / escalates

NO relevant information → Communicate that information was not found → Existing support process / Escalation

Retrieval or AI service unavailable → Service/fallback message → Existing support process / Escalation

15.5 Knowledge Conflict Handling

When multiple approved sources provide conflicting information, the assistant should not automatically assume that the most recently retrieved or highest-ranked result is correct.

The system should identify the conflict where possible and provide the available source references to the support agent for verification.

The identified conflict can also be reported to the Knowledge Management Team so that the underlying documentation can be reviewed and updated.

15.6 Continuous Improvement from Edge Cases

Edge cases should contribute to ongoing product and knowledge-base improvement.

The following information can be monitored:

- Frequently unanswered queries.
- Frequently escalated query types.
- Queries producing negative agent feedback.
- Retrieval failures.
- Conflicting knowledge sources.
- Outdated or incomplete documentation.
- Repeated requests for clarification.
- AI responses requiring frequent human correction.

Analysis of these patterns can help stakeholders determine whether improvements are required in the knowledge base, retrieval process, response-generation process, user experience, or support workflow.

16. AI Evaluation

16.1 Overview

The effectiveness of the AI Customer Support Assistant should be evaluated using a combination of AI-specific quality measures, operational metrics, and human feedback.

Evaluation should assess not only whether the assistant produces useful responses, but also whether the responses are supported by approved knowledge sources, relevant to the query, complete, and appropriately handle uncertainty.

16.2 Evaluation Dimensions

The proposed AI evaluation framework will assess the following dimensions:

Evaluation Dimension	What It Measures	Example Evaluation Question
Retrieval Relevance	Whether the retrieved knowledge is relevant to the submitted query.	Did the system retrieve information that is useful for answering the query?
Groundedness	Whether the generated response is supported by the retrieved information.	Can the claims in the response be supported by the retrieved sources?
Response Relevance	Whether the response directly addresses the agent's question.	Does the response answer what the agent actually asked?
Response Completeness	Whether important information from the available supporting context is included.	Does the response address the key aspects needed to resolve the query?
Source Accuracy	Whether the sources shown actually support the response.	Do the referenced documents contain information supporting the generated answer?
Factual Accuracy	Whether the response is factually correct based on the approved knowledge available.	Is the information provided consistent with the authoritative source?
Hallucination / Unsupported Content	Whether the assistant introduces information not supported by available sources.	Did the assistant state information that cannot be supported by the retrieved knowledge?
Uncertainty Handling	Whether the assistant appropriately acknowledges insufficient information.	Does the assistant avoid confidently answering when evidence is insufficient?

Safety & Access Compliance	Whether the assistant follows access, security, and human-oversight requirements.	Did the system avoid exposing restricted information or bypassing required human review?
Agent Usefulness	Whether support agents find the response useful in their workflow.	Did the response help the agent resolve the customer's query?

16.3 Evaluation Dataset

A representative evaluation dataset should be created using realistic support queries covering different categories and levels of difficulty.

The dataset may include:

- Frequently asked product questions.
- Common troubleshooting queries.
- Account and configuration-related questions.
- Queries requiring information from multiple documents.
- Follow-up questions requiring conversation context.
- Ambiguous queries.
- Queries where relevant information exists in the knowledge base.
- Queries where relevant information is unavailable.
- Queries involving conflicting documentation.
- Queries outside the assistant's supported scope.

Each evaluation query should have an expected answer or reference information against which the AI-generated response can be assessed.

16.4 Human Evaluation

Because AI response quality cannot always be fully evaluated using automated metrics, human review should form an important part of the evaluation process.

A sample of AI-generated responses can be reviewed by support or subject-matter experts against defined evaluation criteria.

Reviewers can assess whether the response is:

- Relevant.
- Correct.
- Complete.
- Grounded in approved sources.
- Clear and understandable.
- Appropriate for the query.
- Properly handling uncertainty when information is insufficient.

- Human evaluation results can be used alongside automated measurements and agent feedback.

16.5 Agent Feedback

Support agents can provide feedback during normal use of the assistant.

The proposed feedback mechanism includes:

- Helpful.
- Not helpful.
- Inaccurate.
- Irrelevant.
- Incomplete.
- Other feedback.

Feedback should be analyzed periodically to identify recurring problems rather than treating an individual feedback response as a definitive measure of AI quality.

16.6 Evaluation of Retrieval

The retrieval component should be evaluated independently from response generation.

The evaluation should determine whether the system is successfully retrieving the information needed by the LLM.

Key areas include:

- Relevance of retrieved content.
- Coverage of relevant knowledge.
- Retrieval of appropriate document sections.
- Retrieval consistency across similar queries.
- Correct application of access restrictions.
- Handling of queries with no relevant information.

Poor retrieval quality can lead to poor AI responses even when the underlying language model is capable of generating a good response.

16.7 Evaluation of Generated Responses

Generated responses should be evaluated against the retrieved context and authoritative knowledge sources.

The evaluation should determine:

1. **Is the response supported?**
The response should not introduce unsupported claims as factual information.
2. **Is the response relevant?**
The response should directly address the submitted query.
3. **Is the response complete?**
Important information needed to answer the query should not be unnecessarily omitted.
4. **Is the response clear?**
The response should be understandable to a support agent.
5. **Are the sources appropriate?**
The displayed sources should support the response.
6. **Is uncertainty handled appropriately?**
The assistant should acknowledge when available information is insufficient.

16.8 Evaluation of Fallback Behavior

The assistant should also be evaluated on how it behaves when it cannot provide a reliable answer.

Test scenarios should include:

- No relevant knowledge found.
- Insufficient retrieved information.
- Conflicting sources.
- Ambiguous query.
- Retrieval failure.
- AI generation failure.
- Restricted information requested.
- Query outside supported scope.

The expected behavior in these scenarios is to communicate limitations clearly and direct the agent toward verification, existing support resources, or escalation where appropriate.

16.9 Evaluation Process

The proposed evaluation process is:

Create Evaluation Dataset



Run Queries Through Assistant



Evaluate Retrieval Quality



Evaluate Generated Response



Review Source Grounding



Assess Safety & Fallback Behavior



Collect Human / Agent Feedback



Identify Recurring Issues



Improve Knowledge / Retrieval / Response Process



Re-evaluate

16.10 Continuous Evaluation

AI evaluation should not be treated as a one-time activity.

The solution should be periodically evaluated as:

- Knowledge sources change.
- Product functionality changes.
- New types of support queries appear.
- Retrieval behavior changes.
- AI models or configurations change.
- Agent feedback identifies new issues.

Recurring evaluation can help determine whether the assistant continues to meet the intended business and user requirements.

16.11 Evaluation Success Criteria

The solution should demonstrate improvement across multiple dimensions rather than being judged using a single AI-quality metric.

Evaluation should consider:

- Quality and relevance of retrieved information.
- Groundedness of generated responses.
- Response relevance and completeness.
- Source-reference accuracy.
- Appropriate uncertainty and fallback behavior.
- Agent feedback.
- Support workflow outcomes.
- Frequency and nature of escalations.

Specific numerical targets can be defined during a future pilot or implementation phase after establishing a baseline using real support data.

17. Product KPIs & Success Metrics

17.1 Overview

The success of the AI Customer Support Assistant should be measured using a combination of business, operational, user, and AI-performance metrics.

The metrics should determine whether the solution reduces support-agent effort, improves response efficiency, supports consistent information delivery, and provides useful AI assistance without compromising human oversight or information quality.

Since this is a self-initiated case study, specific numerical targets are not assumed. Baseline values and target improvements should be established during a pilot using actual support data.

17.2 KPI Framework

KPI ID	KPI	What It Measures	Related Business Objective
KPI-001	Average Knowledge Search Time	Average time an agent spends finding information needed to address a query.	BO-001
KPI-002	Average Response Time	Time taken by agents to provide an initial response after receiving a customer query.	BO-002
KPI-003	Average Resolution Time	Average time required to resolve a customer query.	BO-002
KPI-004	Routine Query Escalation Rate	Percentage of routine queries escalated to senior/higher-level support.	BO-004
KPI-005	AI Assistant Adoption Rate	Percentage of eligible support agents actively using the assistant.	BO-005
KPI-006	Query Assistance Rate	Percentage of supported queries for which agents use the assistant.	BO-005
KPI-007	Agent Helpfulness Rate	Percentage of agent feedback indicating that the AI response was helpful.	BO-005
KPI-008	Response Groundedness Rate	Percentage of evaluated responses adequately supported by retrieved approved information.	BO-003
KPI-009	Source Reference Accuracy	Percentage of evaluated responses where displayed sources appropriately support the generated response.	BO-003
KPI-010	Unsupported Response Rate	Percentage of evaluated responses containing information that cannot be adequately supported by available sources.	BO-003
KPI-011	Knowledge Gap Rate	Frequency of queries where relevant information is unavailable or insufficient in the approved knowledge base.	BO-003 / BO-005

KPI-012	Fallback / Escalation Rate	Percentage of queries where the assistant cannot reliably provide an answer and requires fallback or escalation.	BO-004
KPI-013	Agent Verification Rate	Percentage of AI-assisted responses reviewed by agents before customer communication.	BO-003 / BO-005
KPI-014	Repeat Query Rate	Frequency of repeated or follow-up support queries that may indicate unresolved issues.	BO-002 / BO-005

17.3 Business Outcome Metrics

The primary business outcomes should focus on whether the assistant improves the support workflow.

17.3.1 Knowledge Search Time

Definition: Average time spent by support agents searching approved knowledge sources before obtaining the information required to address a query.

Purpose: Determine whether the assistant reduces the effort associated with searching across multiple knowledge sources.

Expected direction: Decrease, compared with the established baseline.

17.3.2 Response Time

Definition: Average time between a customer query being received and the support agent providing an initial response.

Purpose: Determine whether faster access to relevant information contributes to faster customer responses.

Expected direction: Decrease, compared with the established baseline.

17.3.3 Resolution Time

Definition: Average time required to resolve a customer support query.

Purpose: Determine whether the assistant contributes to more efficient resolution of supported queries.

Expected direction: Decrease, where the assistant is relevant to the query type.

17.3.4 Routine Query Escalation Rate

Definition: Percentage of routine support queries escalated to senior or specialized support teams.

Purpose: Determine whether agents can resolve a greater proportion of routine queries using available information.

Expected direction: Decrease, provided resolution quality is maintained.

17.4 User / Agent Metrics

The assistant's success also depends on whether support agents actually find it useful and incorporate it into their workflow.

Key measures include:

- **AI Assistant Adoption Rate:** Percentage of eligible agents actively using the assistant.
- **Query Assistance Rate:** Percentage of supported queries where the assistant is used.
- **Agent Helpfulness Rate:** Percentage of responses marked helpful by agents.
- **Agent Verification Rate:** Percentage of AI-generated responses reviewed before customer communication.
- **Feedback Rate:** Percentage of AI-assisted interactions for which agents provide feedback.

Low adoption should not automatically be interpreted as a product failure. It should be investigated alongside qualitative feedback, workflow fit, response quality, usability, and other operational factors.

17.5 AI Quality Metrics

AI-specific metrics should be monitored alongside business outcomes.

Key measures include:

- **Response Groundedness Rate:** Whether generated responses are adequately supported by retrieved information.
- **Source Reference Accuracy:** Whether displayed sources support the claims made in the response.
- **Unsupported Response Rate:** Frequency with which the assistant provides unsupported information.
- **Retrieval Relevance:** Whether retrieved information is relevant to the submitted query.
- **Response Relevance:** Whether the generated response directly addresses the query.

- **Response Completeness:** Whether important supporting information is appropriately included.
- **Fallback Rate:** Frequency with which the assistant correctly identifies that sufficient information is unavailable.

17.6 Knowledge Base Metrics

Because the quality of a RAG-based assistant depends significantly on its knowledge sources, the knowledge base should also be monitored.

Relevant metrics include:

- Frequency of unanswered queries.
- Frequency of queries associated with knowledge gaps.
- Number of conflicting knowledge sources identified.
- Frequency of outdated knowledge being identified.
- Most frequently referenced knowledge sources.
- Most frequently escalated query categories.

These measures can help the Knowledge Management Team prioritize documentation improvements.

17.7 KPI Relationship to Business Objectives

The relationship between business objectives and KPIs can be summarized as follows:

Business Objective	Primary Metrics
BO-001 — Reduce time spent searching for information	KPI-001, KPI-006
BO-002 — Improve response speed	KPI-002, KPI-003
BO-003 — Improve consistency and accuracy	KPI-008, KPI-009, KPI-010, KPI-013
BO-004 — Reduce unnecessary escalation	KPI-004, KPI-012
BO-005 — Improve agent productivity and customer experience	KPI-005, KPI-007, KPI-003, KPI-014

17.8 Measurement Approach

The recommended measurement approach is to establish a baseline before evaluating the impact of the assistant.

Baseline Measurement

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Pilot / Controlled Rollout

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Collect Usage + AI Quality + Operational Metrics

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Compare Against Baseline

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Analyze Agent Feedback

↓

Identify Improvement Areas

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Iterate Product / Knowledge Base

↓

Re-measure

Where possible, metrics should be segmented by query type because the assistant may be more useful for some support scenarios than others.

For example, routine product-usage queries may be evaluated separately from complex technical or account-specific queries.

17.9 Success Measurement Principles

The solution should not be considered successful based on AI usage or response speed alone.

Success should be evaluated by considering:

- Whether support-agent effort is reduced.
- Whether customer response and resolution processes become more efficient.
- Whether responses remain grounded in approved information.
- Whether agents find the assistant useful.
- Whether unnecessary escalation decreases without compromising response quality.
- Whether the assistant appropriately handles uncertainty and unsupported queries.
- Whether recurring knowledge gaps can be identified and addressed.

Specific numerical targets should be established after baseline measurement and during a pilot or implementation phase.

18. Requirements Traceability Matrix

18.1 Overview

The Requirements Traceability Matrix (RTM) establishes a clear relationship between business objectives, business requirements, functional requirements, user stories, and acceptance criteria.

The purpose of the RTM is to ensure that requirements remain aligned with the intended business outcomes and that each requirement can be traced through the proposed solution.

The RTM also supports requirements validation, impact analysis, testing, and future change management.

18.2 Traceability Matrix

Business Objective	Business Requirement	Functional Requirement	User Story	Acceptance Criteria Reference
BO-001 — Reduce time spent searching across knowledge sources	BR-001 — Provide centralized natural-language search across approved sources	FR-001, FR-004	US-001, US-002	US-001 AC, US-002 AC
BO-001 — Reduce time spent searching across knowledge sources	BR-002 — Quickly identify relevant information for customer queries	FR-004, FR-005, FR-009	US-002, US-003	US-002 AC, US-003 AC
BO-002 — Improve speed of responding to customer queries	BR-003 — Provide AI-assisted responses based on retrieved information	FR-008, FR-009, FR-010	US-003, US-006	US-003 AC, US-006 AC
BO-003 — Improve consistency and accuracy of information	BR-003 — Provide AI-assisted responses based on retrieved information	FR-008, FR-009, FR-010	US-003, US-009	US-003 AC, US-009 AC
BO-003 — Improve consistency and accuracy of information	BR-004 — Provide supporting source references	FR-013, FR-014, FR-015	US-004	US-004 AC

BO-004 — Reduce unnecessary escalation of routine queries	BR-005 — Reduce unnecessary escalation of routine queries	FR-019, FR-020	US-007	US-007 AC
BO-003 / BO-005 — Improve response quality and support-agent productivity	BR-006 — Allow agents to provide feedback	FR-016, FR-017, FR-018	US-008, US-013	US-008 AC, US-013 AC
BO-003 / BO-004 — Support safe handling of uncertain queries	BR-007 — Provide fallback and escalation when information is insufficient	FR-007, FR-019, FR-020	US-006, US-007, US-017	US-006 AC, US-007 AC, US-017 AC
BO-003 / BO-005 — Maintain reliable and authorized information access	BR-008 — Use only approved and authorized information sources	FR-006, FR-022, FR-023, FR-024	US-010, US-014	US-010 AC, US-014 AC
BO-005 — Improve support-agent productivity and customer experience	BR-009 — Evaluate effectiveness using performance and business metrics	FR-025, FR-026, FR-027	US-011, US-012, US-013	US-011 AC, US-012 AC, US-013 AC
BO-003 / BO-005 — Maintain human oversight	BR-010 — Provide human oversight for uncertain, sensitive, or complex queries	FR-012, FR-020, FR-021	US-007, US-009, US-017	US-007 AC, US-009 AC, US-017 AC

18.3 Traceability Coverage

The RTM provides traceability across the major requirement layers:

Business Objectives

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Business Requirements

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Functional Requirements

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User Stories

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Acceptance Criteria

This structure ensures that the proposed solution remains connected to the original business problem rather than defining features independently of business needs.

18.4 Requirements Validation

Before implementation, stakeholders should review the RTM to confirm that:

- Each business objective is supported by one or more business requirements.
- Each business requirement is supported by relevant functional requirements.
- Functional requirements are represented through user stories where applicable.
- User stories have clear and testable acceptance criteria.
- Requirements do not introduce functionality unrelated to the defined scope.
- Security, fallback, human oversight, and evaluation requirements are represented.
- Changes to requirements can be assessed for their potential impact on related requirements and solution components.

Appendix A — Glossary

Term	Definition
AI / Artificial Intelligence	Technology that enables systems to perform tasks that normally require human intelligence, such as understanding questions and generating responses.
LLM / Large Language Model	An AI model capable of understanding and generating natural-language text.
RAG / Retrieval-Augmented Generation	An approach where relevant information is retrieved from approved knowledge sources and provided to an LLM to help generate a grounded response.
Knowledge Source	An approved document or content source that contains information used to support customer support activities.
Knowledge Base	The searchable representation of approved organizational knowledge used by the retrieval process.

Embedding	A numerical representation of text that captures semantic meaning and can be used to identify related content.
Vector Database / Vector Search Index	A system used to store and search embeddings to identify content that is semantically relevant to a query.
Chunking	The process of dividing documents into smaller sections so that relevant portions can be retrieved efficiently.
Retrieval	The process of finding relevant information from approved knowledge sources based on a submitted query.
Groundedness	The extent to which an AI-generated response is supported by the information retrieved from approved knowledge sources.
Hallucination / Unsupported Content	Information generated by an AI system that is not supported by the available authoritative information.
Fallback	The behavior followed when the assistant cannot reliably provide an answer, such as communicating uncertainty or directing the agent to existing support processes.
Human-in-the-Loop	An approach where a human remains responsible for reviewing, validating, or making decisions based on AI-assisted output.
Knowledge Gap	A situation where the approved knowledge sources do not contain sufficient information to address a support query.